Solitons in arbitrary dimensions stabilized by photon mediated interactions: Supplemental Materials

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S1. SYSTEM

Here we provide a detailed derivation of the wave-packet dynamics, in terms of a spin model, following a description recently derived in Ref. [S1]. The initial state consists of a wave packet with average momentum p_0 and width σ_p prepared from an array of thermal atoms via velocity selection— implemented by a Raman pulse—.

The preparation step finished with a two-photon Bragg transition, that prepares a superposition between $|\Downarrow_p\rangle_n \equiv |p_0 + p\rangle_n$ and $|\Uparrow_p\rangle_n \equiv |p_0 + p + 2\hbar k\rangle_n$, with k being the wave vector of the cavity model and $\sigma_p \ll 2\hbar k$.

The numerical simulations are done in the following steps:

- Omitting other labels irrelevant to the dynamics, we focus on a specific experimental realization with an initial set of position and momentum states labeled as $p_1, p_2, \ldots, p_N$, sampled from a Gaussian distribution of width σ_p, within the range $[p_0 - 2\hbar k, p_0]$. Specifically, we divide the range $[p_0 - 2\hbar k, p_0]$ into L discrete sites with spacing $\delta p = 2\hbar k/L$, and generate a set of momentum states sampled for the Gaussian distribution. The we apply a Bragg pulse to generate coherence.
- We the numerically integrate the coupled non-linear equations Eq. (7) below using the Runge-Kutta method to obtain the wave function at each p_n for atom n over time.
- We repeat the above procedure for different realizations of initial momentum configurations. In our simulations, we set $L = 500$ and perform 10^4 samples to ensure convergence.
- Using them we compute the discrete Fourier transform to obtain the real-space wave function.

A. Spin model

As explained in the main text, we define the pseudospin-1/2 operators written in terms of annihilation operators $\hat{\psi}^{\Downarrow}_{p_n}$ for $|\Downarrow_p\rangle_n$ momentum state and $\hat{\psi}^{\Uparrow}_{p_n}$ for $|\Uparrow_p\rangle_n$ momentum state. They are given by

$$\begin{aligned}
\hat{s}^X_{p_n} &= \frac{1}{2}\left((\hat{\psi}^{\Uparrow}_{p_n})^\dagger \hat{\psi}^{\Downarrow}_{p_n} + (\hat{\psi}^{\Downarrow}_{p_n})^\dagger \hat{\psi}^{\Uparrow}_{p_n}\right), \\
\hat{s}^Y_{p_n} &= \frac{1}{2i}\left((\hat{\psi}^{\Uparrow}_{p_n})^\dagger \hat{\psi}^{\Downarrow}_{p_n} - (\hat{\psi}^{\Downarrow}_{p_n})^\dagger \hat{\psi}^{\Uparrow}_{p_n}\right), \qquad \text{(S1)}\\
\hat{s}^Z_{p_n} &= \frac{1}{2}\left((\hat{\psi}^{p_n}_{\Uparrow})^\dagger \hat{\psi}^{p_n}_{\Uparrow} - (\hat{\psi}^{p_n}_{\Downarrow})^\dagger \hat{\psi}^{p_n}_{\Downarrow}\right),
\end{aligned}$$

In terms of them we define collective spin operators

$$\hat{S}_\alpha = \sum_n \hat{s}^\alpha_{p_n} \quad \alpha \in [X, Y, Z, +, -]. \tag{S2}$$

After adiabatically eliminating the excited state and the cavity field, we obtain an effective Hamiltonian in the ground state manifold [S1], evolving only the motional degree of freedom. It is given by:

$$\hat{H} = \hat{H}_{\rm ex} + \hat{H}_{\rm in} \tag{S3}$$

The exchange interaction can be written as:

$$\hat{H}_{\rm ex} = \chi \hat{S}_+ \hat{S}_- = \chi\left(\hat{S}^2 - \hat{S}_Z^2 + \hat{S}_Z\right), \tag{S4}$$

and the single particle dispersion terms as ,

$$\begin{aligned}\hat{H}_{\mathrm{in}}/\hbar &= \sum_n \omega^{\Downarrow}_{p_n} (\hat{\psi}^{\Downarrow}_{p_n})^\dagger \hat{\psi}^{\Downarrow}_{p_n} + \omega^{\Uparrow}_{p_n} (\hat{\psi}^{\Uparrow}_{p_n})^\dagger \hat{\psi}^{\Uparrow}_{p_n} \\ &= \sum_n \left(\omega_Z + \frac{2kp_n}{M}\right) \hat{s}^Z_{p_n} + \Big[\frac{p_n^2}{2M\hbar} + \frac{(p_0 + \hbar k_c)\, p_n}{M\hbar}\Big] \Big[(\hat{\psi}^{\Downarrow}_{p_n})^\dagger \hat{\psi}^{\Downarrow}_{p_n} + (\hat{\psi}^{\Uparrow}_{p_n})^\dagger \hat{\psi}^{\Uparrow}_{p_n}\Big],\end{aligned} \tag{S5}$$

Here M is the atom's mass. The single-particle energies above are given by

$$\begin{aligned}\hbar\omega^{\Downarrow}_p &= \frac{(p_0 + p)^2}{2M} = \frac{p_0^2}{2M} + \frac{p_0 p}{M} + \frac{p^2}{2M} \\ \hbar\omega^{\Uparrow}_p &= \frac{(p_0 + 2\hbar k_c + p)^2}{2M} = \frac{(p_0 + 2\hbar k_c)^2}{2M} + \frac{(p_0 + 2\hbar k_c)\, p}{M} + \frac{p^2}{2M}\end{aligned} \tag{S6}$$

and $\hbar\omega_Z = (p_0 + 2\hbar k_c)^2 /2M - p_0^2/2M$.

B. Mean-field Approximation

In this part, we derive the mean-field equations of motion for the field operator $\hat{\psi}^{\Uparrow(\Downarrow)}_{p_n}$, by approximating it as a c-number, $\psi^{\Uparrow(\Downarrow)}_{p_n} \approx \left\langle \hat{\psi}^{\Uparrow(\Downarrow)}_{p_n} \right\rangle$. They are given by,

$$\begin{aligned}i\frac{d}{dt}\psi^{\Downarrow}_{p_n} &= \omega^{\Downarrow}_{p_n} \psi^{\Downarrow}_{p_n} + \chi\psi^{\Uparrow}_{p_n} \sum_m (\psi^{\Uparrow}_{p_m})^* \psi^{\Downarrow}_{p_m} \\ i\frac{d}{dt}\psi^{\Uparrow}_{p_n} &= \omega^{\Uparrow}_{p_n} \psi^{\Uparrow}_{p_n} + \chi\psi^{\Downarrow}_{p_n} \sum_m (\psi^{\Downarrow}_{p_m})^* \psi^{\Uparrow}_{p_m}\end{aligned} \tag{S7}$$

with $|\psi^{\Downarrow}_{p_n}|^2 + |\psi^{\Uparrow}_{p_n}|^2 = 1$ and $\psi^{\Downarrow}_{p_n} = \psi^{\Uparrow}_{p_n} = \frac{1}{\sqrt{2}}$ for the initial state, for the case of a $\pi/2$ pulse.

In the limit where $\chi N \gg 2k\sigma_p/m$, atoms with different momentum lock together, $\left\langle \hat{S}_X \right\rangle \approx N/2$ and the exchange interaction simplifies to $\hat{H}_{\mathrm{ex}} = \chi N \hat{S}_X$. The exchange interactions acts effectively as a strong transverse field along the initial spin direction. The corresponding equations of motion become:

$$\begin{aligned}i\frac{d}{dt}\psi^{\Downarrow}_{p_n} &= \omega^{\Downarrow}_p \psi^{\Downarrow}_{p_n} + \frac{\chi N}{2}\psi^{\Uparrow}_{p_n} \\ i\frac{d}{dt}\psi^{\Uparrow}_{p_n} &= \omega^{\Uparrow}_p \psi^{\Uparrow}_{p_n} + \frac{\chi N}{2}\psi^{\Downarrow}_{p_n}.\end{aligned} \tag{S8}$$

After evolution under $\hat{H}$ for time t_d, and replacing $p_n \to p$ for simplicity we get:

$$\begin{aligned}\sqrt{2}\psi^{\Downarrow}_p &= e^{-i\frac{p^2}{2M\hbar}t_d}\Big[\cos\left(\frac{t_d}{2}\sqrt{(2kp/M)^2 + (\chi N)^2}\right) + \frac{i\,(2kp/M - \chi N)\sin\left(\frac{t_d}{2}\sqrt{(2kp/M)^2 + (\chi N)^2}\right)}{\sqrt{(2kp/M)^2 + (\chi N)^2}}\Big] \\ \sqrt{2}\psi^{\Uparrow}_p &= e^{-i\frac{p^2}{2M\hbar}t_d}\Big[\cos\left(\frac{t_d}{2}\sqrt{(2kp/M)^2 + (\chi N)^2}\right) - \frac{i\,(2kp/M + \chi N)\sin\left(\frac{t_d}{2}\sqrt{(2kp/M)^2 + (\chi N)^2}\right)}{\sqrt{(2kp/M)^2 + (\chi N)^2}}\Big].\end{aligned} \tag{S9}$$

Using the approximation

$$\frac{2kp/M \pm \chi N}{\sqrt{(2kp/M)^2 + (\chi N)^2}} \approx \pm\mathrm{sgn}(\chi), \tag{S10}$$

one obtains

$$\sqrt{2}\psi_p^{\Downarrow} \approx \sqrt{2}\psi_p^{\Uparrow} \approx e^{-i\frac{p^2}{2M\hbar}t_d} e^{-i\frac{t_d}{2}\sqrt{\left(\frac{2k_c p}{M}\right)^2+(\chi N)^2}} \approx e^{-i\frac{\chi N}{2}t_d} e^{-i\left(\frac{1}{2M\hbar}+\frac{k_c^2}{\chi N M^2}\right)p^2 t_d}. \tag{S11}$$

At the optimal interaction strength, χ_{opt},

$$\chi_{\mathrm{opt}} N = -4\frac{\hbar k^2}{2M} = -4E_R. \tag{S12}$$

the global phase is canceled,

C. General initial polar angle

In this section, we consider the case where the initial state is prepared as a superposition between $|p_0\rangle$ and $|p_0 + 2\hbar k\rangle$ with an arbitrary polar angle θ, as described in the main text. In this case, the conserved total magnetization $\left\langle \hat{S}_Z \right\rangle = \frac{N}{2}\cos\theta$. Under the mean-field approximation and assuming dominant exchange interactions over single-particle inhomogeneities $\left\langle \hat{S}_+ \right\rangle \approx \frac{N}{2}\sin\theta e^{-2i\chi\langle \hat{S}_Z\rangle t}$ [S2].

The mean-field Hamiltonian thus acquires a time-dependent form:

$$\hat{H}_{\mathrm{MF}}/\hbar = \hat{H}_{\mathrm{in}}^d/\hbar + \frac{\chi N \sin\theta}{2}\left(\hat{S}_+ e^{i\chi N\cos\theta t} + \hat{S}_- e^{-i\chi N\cos\theta t}\right). \tag{S13}$$

By transforming into a rotating frame defined by $\hat{H}_R = -\chi N\cos\theta \hat{S}_Z$, the Hamiltonian simplifies to:

$$\hat{H}'_{\mathrm{MF}}/\hbar = \sum_r \left(\frac{2kp_r}{M} + \chi N\cos\theta\right)\hat{s}^Z_{p_r} + \chi N\sin\theta \hat{S}_X. \tag{S14}$$

The relevant eigenstate in this rotating frame has the energy dispersion:

$$\begin{aligned} E_p &= -\frac{1}{2}\sqrt{\left(\chi N\cos\theta + \frac{2kp}{M}\right)^2 + (\chi N\sin\theta)^2} + \frac{p^2}{2M\hbar} \\ &\approx \frac{\chi N}{2} - \frac{kp\cos\theta}{M} + \frac{p^2}{2M\hbar}\left(1 + \frac{4E_R\sin^2\theta}{\chi N}\right). \end{aligned} \tag{S15}$$

The linear term in p generates a shift in collective recoil of the two wave packets, by $-\frac{k\cos\theta}{M}$ and thus a net average momentum of $p_0 + \hbar k(1-\cos\theta)$. Then the optimal interaction strength that cancels the dispersion is given by

$$\chi_{\mathrm{opt}} N = -4E_R\sin^2\theta. \tag{S16}$$

S2. DETECTION SCHEME FOR THERMAL SOLITONS

We now discuss the detection scheme proposed in the main text for thermal solitons. In the following, we focus primarily on the 1D scenario. Soliton generation in 2D and 3D systems already requires the introduction of additional internal degrees of freedom to enhance the effective energy difference ω_Z, and therefore adding another state is possible but more cumbersome. In the high-dimensional cases, therefore, it will be easier to start with a BEC and use standard time of flight images.

The protocol relies on the fact that a similar flattening of the band can be realized for non-interacting atoms using an additional driving field along the x-axis of the Bloch sphere with Rabi frequency $\Omega = -4E_R$. As illustrated in Fig. S1(a,ii), the key idea then is to use an additional external drive (red arrows), instead of exchange interactions, to prepare a soliton between a pair of momentum states in an other hyperfine level $|\uparrow\rangle$, and inject it into one arm of the interferometer. We choose the cavity frequency to be off-resonant from the atomic transition for this internal state, making it insensitive to exchange interactions. In the other arm, we inject a soliton in a hyperfine state $|\downarrow\rangle$ that experiences exchange interactions (orange arrows). The contrast achieved at the output of the interferometer serves as a probe of the differential dispersion.

We adopt the notation $|\Downarrow_p, \Uparrow_p; \downarrow, \uparrow\rangle$ for the momentum (first index) and spin basis (second index), and ignore the atom number index r for simplicity. The interferometer sequence, shown in the spacetime diagram of Fig. S1(b), consists of the following steps:

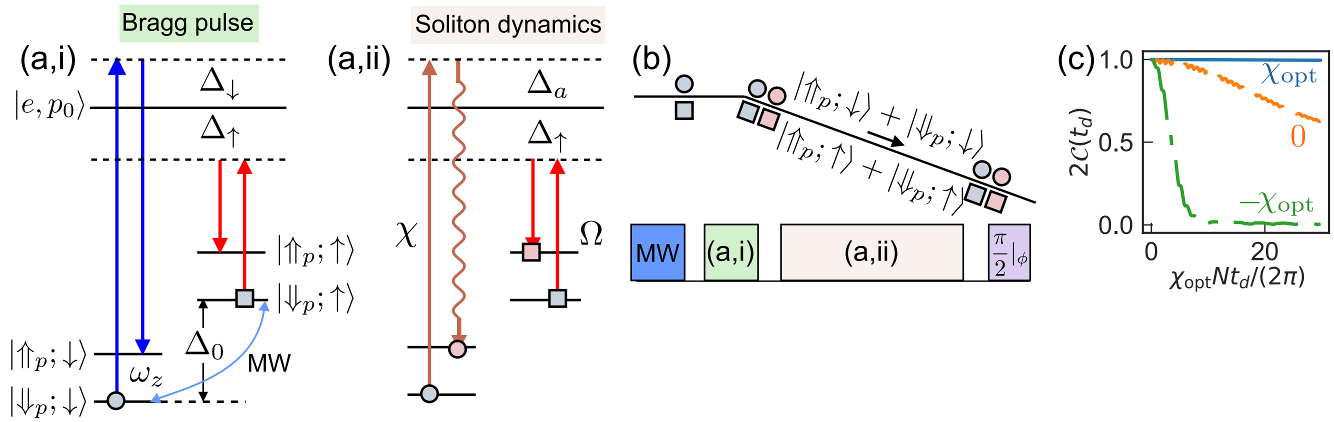

FIG. S1. Detection scheme for a thermal cloud (a,i) A microwave drive couples two hyperfine levels with an energy difference Δ_0 (light blue arrow). Two sets of Bragg beams (blue and red) independently prepare momentum superpositions in each of the hyperfine levels.(a,ii) Cavity photons (orange) and Bragg beams (red) are used to control the motional dynamics for different hyperfine levels respectively. (b) Interferometer sequence to measure wave-packet broadening. (c) Simulated measurement outcome using a Bragg pulse with Rabi frequencies, $\Omega = \chi_{\rm opt}$ (red), $\chi = \chi_{\rm opt}$ (blue), $\chi = -\chi_{\rm opt}$ (orange) and $\chi = 0$ (green).

- After velocity selection $|\Downarrow_p, \downarrow\rangle$, a microwave pulse is used to create a superposition of internal states $\frac{1}{\sqrt{2}}(|\Downarrow_p, \downarrow\rangle + |\Downarrow_p, \uparrow\rangle)$ (Fig. S1(a,i) light blue arrow);
- Two sets of Bragg beams with opposite single-photon detunings $\Delta_{\downarrow(\uparrow)}$ from the excited state (Fig. S1(a,i) dark blue and red arrows) generate a density grating within each internal state as $\frac{1}{2}(|\Uparrow_p, \downarrow\rangle + |\Downarrow_p, \downarrow\rangle + |\Uparrow_p, \uparrow\rangle + |\Downarrow_p, \uparrow\rangle)$;
- The blue-detuned Bragg beams are turned off and the red-detuned Bragg frequency is tuned to $\Omega = \chi_{\rm opt}$ (red) for the hyperfine level $|\uparrow\rangle$, while the cavity-mediated interactions generated by the probe beam (orange) are experienced by the hyperfine level $|\downarrow\rangle$ (Fig. S1(a,ii));
- After some time evolution, a second set of Raman and a microwave $\pi/2$ pulses is applied. The fringe contrast reveals the overlap between the wave packets in the two arms.

In this way, the spatial overlap between wave packets in the interferometer arms can be mapped onto the population difference at the interferometer output.

Figure . S1(c)) shows an example of the signal performed by varying the exchange interaction strength χN for $|\downarrow\rangle$ while setting the Raman drive Rabi frequency to $\Omega = \chi_{\rm opt}$: Three cases are used, $\chi_{\rm opt}$ (blue), 0 (green), and $-\chi_{\rm opt}$ (orange). The simulations reveal distinct behaviors in each case: for $\chi = \chi_{\rm opt}$, both of the $|\uparrow\rangle$ and $|\downarrow\rangle$ momentum superpositions exhibit perfect soliton dynamics, maintaining constant overlap; with $\chi = 0$, the wave packets for $|\downarrow\rangle$ spatially separate due to lack of gap protection; in this case an echo sequence should help to increase the overlap but will not be perfect due to broadening of one of the superpositions. For $\chi = -\chi_{\rm opt}$, the wave packets for $|\downarrow\rangle$ remain bound but broaden, and the contrast can decay faster than the non-interacting case.

S3. BALANCED PUMP SCHEME

In the prior discussions, we focused on ideal unitary dynamics that excluded dissipation. However, Ref. [S1, S3] demonstrated that superradiance is a detrimental effect, that can drive the state toward either the north or south pole, disrupting coherence and reducing the gap protection.

A straightforward strategy to mitigate superradiance is to detune the pump field further from the cavity resonance. However, this approach results in slower dynamics that can be then significantly affected by light scattering. An alternative method is illustrated in Fig. S2, where the cavity is driven by two distinct frequency tones. Unlike the dual-pump resonance condition in [S3, S4], here we choose $\Delta_1 \neq \Delta_2$. In this setup, the two pumps independently

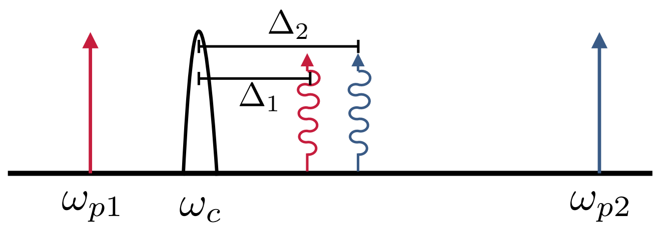

FIG. S2. A two pump scheme can be used to cancel superradiance and maintain a dominant unitary dynamics.

generate exchange interactions, described by:

$$\begin{aligned}\hat{H} &= \chi_1 \hat{S}_- \hat{S}_+ + \chi_2 \hat{S}_+ \hat{S}_- \approx (\chi_1 + \chi_2)\left(\hat{S}^2 - (\hat{S}_Z)^2\right),\\ \hat{L}_1 &= \sqrt{\Gamma_1}\hat{S}_+, \quad \hat{L}_2 = \sqrt{\Gamma_2}\hat{S}_-,\end{aligned} \tag{S17}$$

and

$$\Gamma_1 = \left(\frac{g^2}{4\Delta_0}\right)^2 |\alpha_1|^2 \frac{\kappa}{\Delta_1^2 + (\kappa/2)^2} \quad \Gamma_2 = \left(\frac{g^2}{4\Delta_0}\right)^2 |\alpha_2|^2 \frac{\kappa}{\Delta_2^2 + (\kappa/2)^2}. \tag{S18}$$

By setting $\Gamma = \Gamma_1 = \Gamma_2$, it is possible to suppress superradiance at the mean-field level. To be more concrete, the mean-field equation of motion get modified in the balanced pump case to

$$\begin{aligned}\frac{d}{dt}\left\langle \hat{S}_X \right\rangle &= \cdots - \Gamma \left\langle \hat{S}_X \right\rangle\\ \frac{d}{dt}\left\langle \hat{S}_Y \right\rangle &= \cdots - \Gamma \left\langle \hat{S}_Y \right\rangle\\ \frac{d}{dt}\left\langle \hat{S}_Z \right\rangle &= \cdots - 2\Gamma \left\langle \hat{S}_Z \right\rangle\end{aligned} \tag{S19}$$

with $\cdots$ representing the contributions from the unitary dynamics. Note that due to the balance in the superradiance processes, the collective enhancement for superradiance disappears, resulting in a much slower time scale (by $O(N)$) compared to the collective dynamics.

S4. EXACT DYNAMICS

In Sec. IV of the Supplementary Material of Ref. [S1], we analyzed the exact dynamics of a *Gedanken* experiment involving a single excitation. Building on these foundations, we now consider a scenario in which the initial state is aligned along the x-direction and derive an analytical expression for the wave packet dynamics. Our results confirm that the optimal interaction strength χ_{opt}, identified via mean-field analysis above, effectively suppresses both the broadening and the separation of the wave packets, as presented in Eq. (S49).

A. Notation

First, we define the creation operators for single-particle wave packets with a Gaussian envelope $\phi(p)$, centered at momenta $p_0 - \hbar k$ and $p_0 + \hbar k$, respectively:

$$\hat{A} = \sum_p \phi(p)\hat{\psi}_p^{\Downarrow}, \quad \hat{B} = \sum_p \phi(p)\hat{\psi}_p^{\Uparrow}, \tag{S20}$$

which satisfy the commutation relations:

$$\left[\hat{A}, \hat{A}^\dagger\right] = \left[\hat{B}, \hat{B}^\dagger\right] = 1, \tag{S21}$$

and

$$\left[\hat{\psi}_p^{\Downarrow}, \hat{A}^\dagger\right] = \left[\hat{\psi}_p^{\Uparrow}, \hat{B}^\dagger\right] = \phi(p), \quad \left[\hat{\psi}_p^{\Downarrow}, (\hat{A}^\dagger)^n\right] = n\phi(p)(\hat{A}^\dagger)^{n-1}, \quad \left[\hat{\psi}_p^{\Uparrow}, (\hat{B}^\dagger)^n\right] = n\phi(p)(\hat{B}^\dagger)^{n-1}. \tag{S22}$$

Then we define the Dicke state as

$$|N_a, N_b\rangle = \frac{(\hat{A}^\dagger)^{N_a}}{\sqrt{N_a!}} \frac{(\hat{B}^\dagger)^{N_b}}{\sqrt{N_b!}} |vac\rangle . \tag{S23}$$

From the commutation relations in Eq. (S22), we find:

$$\begin{aligned} \hat{\psi}_p^{\Downarrow}\left|\frac{N}{2} - m, \frac{N}{2} + m\right\rangle &= \phi(p)\sqrt{\frac{N}{2} - m}\left|\frac{N}{2} - m - 1, \frac{N}{2} + m\right\rangle, \\ \hat{\psi}_p^{\Uparrow}\left|\frac{N}{2} - m, \frac{N}{2} + m\right\rangle &= \phi(p)\sqrt{\frac{N}{2} + m}\left|\frac{N}{2} - m, \frac{N}{2} + m - 1\right\rangle. \end{aligned} \tag{S24}$$

The initial state of the system, after applying a $\pi/2$ pulse along the y-axis to $|N, 0\rangle$, is given in the Dicke basis by

$$|\psi\rangle_0 = \frac{1}{2^{\frac{N}{2}}} \sum_{m=-\frac{N}{2}}^{\frac{N}{2}} \begin{pmatrix} N \\ \frac{N}{2} + m \end{pmatrix}^{1/2} \left|\frac{N}{2} - m, \frac{N}{2} + m\right\rangle, \tag{S25}$$

The observable of interest is the real-space density $\hat{c}_z^\dagger \hat{c}_z$, where the creation operator at position z is defined as

$$\hat{c}_z^\dagger \propto \sum_p (\hat{\psi}_p^{\Downarrow})^\dagger e^{-i(p_0 - \hbar k + p)z/\hbar} + \sum_p (\hat{\psi}_p^{\Uparrow})^\dagger e^{-i(p_0 + \hbar k + p)z/\hbar}. \tag{S26}$$

We aim to evaluate

$$\hat{c}_z^\dagger \hat{c}_z \propto \sum_{pq} e^{i\omega_{pq}t} \left[(\hat{\psi}_p^{\Downarrow})^\dagger \hat{\psi}_q^{\Downarrow} + (\hat{\psi}_p^{\Uparrow})^\dagger \hat{\psi}_p^{\Uparrow} + (\hat{\psi}_p^{\Downarrow})^\dagger \hat{\psi}_q^{\Uparrow} e^{i(2k_c z - \omega_Z t)} + (\hat{\psi}_p^{\Uparrow})^\dagger \hat{\psi}_p^{\Downarrow} e^{i(\omega_Z t - 2k_c z)}\right] e^{i(q-p)z/\hbar}, \tag{S27}$$

where the global phase ω_{pq} is given by $\omega_{pq} = p_0\,(p - q)\,/m + \left(p^2 - q^2\right)/2m\hbar$. The first terms leads to the center of mass motion at momenta p_0 and second term for the broadening of the wave packets.

B. Derivation

In this section, we derive the wave packet dynamics by evolving operators in the Heisenberg picture:

$$\hat{O}(t) = e^{i\hat{H}t/\hbar}\hat{O}e^{-i\hat{H}t/\hbar} = \hat{O} + i\frac{t}{\hbar}[\hat{H}, \hat{O}] - \frac{(t/\hbar)^2}{2!}[\hat{H}, [\hat{H}, \hat{O}]] + \cdots \tag{S28}$$

We begin by evaluating the commutators between $\hat{\psi}_p^{\Downarrow}$ and $\hat{H}_{\text{ex}}$:

$$\left[\hat{S}_+\hat{S}_-, \hat{\psi}_p^{\Downarrow}\right] = -\hat{S}_+\hat{\psi}_p^{\Uparrow} \quad \left[\hat{S}_+\hat{S}_-, \hat{S}_+\hat{\psi}_p^{\Uparrow}\right] = -\hat{S}_+\hat{S}_-\hat{\psi}_p^{\Downarrow} - (2\hat{S}_Z - 1)\hat{S}_+\hat{\psi}_p^{\Uparrow}. \tag{S29}$$

The n-th order commutator (with coefficient $\frac{(it)^n}{n!}$ in the expansion) takes the form:

$$f^{(n)}(\hat{S}_+\hat{S}_-, 2\hat{S}_Z - 1)\hat{\psi}_p^{\Downarrow} + g^{(n)}(\hat{S}_+\hat{S}_-, 2\hat{S}_Z - 1)\hat{S}_+\hat{\psi}_p^{\Uparrow}, \tag{S30}$$

where $f^{(n)}$ and $g^{(n)}$ are functions of $\hat{S}+\hat{S}-$ and $2\hat{S}_Z - 1$. Furthermore,

$$\begin{aligned} \left[\hat{S}_+\hat{S}_-, f^{(n-1)}\hat{\psi}_p^{\Downarrow} + g^{(n-1)}\hat{S}_+\hat{\psi}_p^{\Uparrow}\right] &= -f^{(n-1)}\hat{S}_+\hat{\psi}_p^{\Uparrow} - g^{(n-1)}[\hat{S}_+\hat{S}_-\hat{\psi}_p^{\Downarrow} + (2\hat{S}_Z - 1)\hat{S}_+\hat{\psi}_p^{\Uparrow}] \\ &= -g^{(n-1)}\hat{S}_+\hat{S}_-\hat{\psi}_p^{\Downarrow} - [f^{(n-1)} + g^{(n-1)}(2\hat{S}_Z - 1)]\hat{S}_+\hat{\psi}_p^{\Uparrow} \\ &= f^{(n)}\hat{\psi}_p^{\Downarrow} + g^{(n)}\hat{S}_+\hat{\psi}_p^{\Uparrow}, \end{aligned} \tag{S31}$$

which leads to the following iterative relation:

$$\begin{pmatrix} f^{(n)} \\ g^{(n)} \end{pmatrix} = -\begin{pmatrix} 0 & \chi\hat{S}_+\hat{S}_- \\ \chi & \chi(2\hat{S}_Z-1) \end{pmatrix}\begin{pmatrix} f^{(n-1)} \\ g^{(n-1)} \end{pmatrix} \equiv M(\hat{S}_+\hat{S}_-, 2\hat{S}_Z-1)\begin{pmatrix} f^{(n-1)} \\ g^{(n-1)} \end{pmatrix}, \tag{S32}$$

allowing us to obtain a closed-form expression for any order of commutator with $\hat{H}_{\rm ex}$.

Next, we consider the commutator with $\hat{H}in$ and add the contribution to $f^{(n)}, g^{(n)}, M$ above:

$$\left[\hat{H}_{in}/\hbar, \hat{\psi}_p^{\Downarrow}\right] = \frac{kp}{M}\hat{\psi}_p^{\Downarrow}, \quad \left[\hat{H}_{in}/\hbar, \hat{S}_+\hat{\psi}_p^{\Uparrow}\right] = -\frac{kp}{M}\hat{S}_+\hat{\psi}_p^{\Uparrow} + \frac{2k}{m}(\sum_q q\hat{s}_q^+)\hat{\psi}_p^{\Uparrow}. \tag{S33}$$

This contributes to $\hat{\psi}_p^{\Downarrow}$ and $\hat{S}_+\hat{\psi}_p^{\Uparrow}$ terms (which still form a closed set under higher-order commutators), along with the unwanted term $\sum_q q\hat{s}_q^+$ (which introduces additional terms).

Below we argue the unwanted term is not important for the later discussion. First we observe

$$\sum_q q\hat{s}_q^+\left|\frac{N}{2}-m, \frac{N}{2}+m\right\rangle = \sqrt{(\frac{N}{2}-m)(\frac{N}{2}+m+1)}\left|\frac{N}{2}-m-1, \frac{N}{2}+m+1\right\rangle \sum_q q\phi(q) = 0, \tag{S34}$$

which apply trivially on the Dicke state. We can then compute the higher-order commutators:

$$\begin{aligned} &[\hat{S}_+\hat{S}_-, (\sum_q q\hat{s}_q^+)\hat{\psi}_p^{\Uparrow}] = -2\hat{S}_+(\sum_q q\hat{s}_q^z)\hat{\psi}_p^{\Uparrow} - (\sum_q q\hat{s}_q^+)\hat{\psi}_p^{\Downarrow}\hat{S}_- \\ &[(\sum_p p\hat{s}_p^z), (\sum_q q\hat{s}_q^+)\hat{\psi}_p^{\Uparrow}] = -p(\sum_q q\hat{s}_q^+)\hat{\psi}_p^{\Uparrow} + 2(\sum_q q^2\hat{s}_q^+)\hat{\psi}_p^{\Uparrow}. \end{aligned} \tag{S35}$$

The first line again vanishes when applying to the Dicke basis, whereas the second line yields a nontrivial contribution:

$$\sum_q q^2\hat{s}_q^+\left|\frac{N}{2}-m, \frac{N}{2}+m\right\rangle = \sqrt{(\frac{N}{2}-m)(\frac{N}{2}+m+1)}\left|\frac{N}{2}-m-1, \frac{N}{2}+m+1\right\rangle \sum_q q^2\phi(q) \approx 0. \tag{S36}$$

These terms are negligible because they are associated with an energy scale $(2k/m)^2\sum_q q^2\phi(q) \sim (2k\sigma_p/m)^2 \ll (E_R/\hbar)^2 \sim \chi_{\rm opt}^2$ within second-order commutators.

As a result, we apply the approximation $\left[\hat{H}_{\rm in}, \hat{S}_+\hat{\psi}_p^{\Uparrow}\right] \approx -\frac{kp}{M}\hat{S}_+\hat{\psi}_p^{\Uparrow}$, and combining this with Eq. (S32) to find:

$$\begin{aligned} \begin{pmatrix} f^{(n)} \\ g^{(n)} \end{pmatrix} &= -\begin{pmatrix} -\frac{kp}{M} & \chi\hat{S}_+\hat{S}_- \\ \chi & \chi(2\hat{S}_Z-1)+\frac{kp}{M} \end{pmatrix}\begin{pmatrix} f^{(n-1)} \\ g^{(n-1)} \end{pmatrix}, \\ &\equiv M(\hat{S}_+\hat{S}_-, 2\hat{S}_Z-1)\begin{pmatrix} f^{(n-1)} \\ g^{(n-1)} \end{pmatrix}. \end{aligned} \tag{S37}$$

With the initial condition $f^{(0)}=1, g^{(0)}=0$, we obtain:

$$\begin{aligned} \hat{\psi}_p^{\Downarrow}(t) &= \sum_{n=0}^{\infty}\frac{(it)^n}{n!}(f^{(n)}\hat{\psi}_p^{\Downarrow} + g^{(n)}\hat{S}_+\hat{\psi}_p^{\Uparrow}) \\ &= e^{-i\frac{(2\hat{S}_Z-1)}{2}}[(\cos\frac{\mathcal{O}}{2}\chi t + i(2\hat{S}_Z-1)\frac{\sin\frac{\mathcal{O}}{2}\chi t}{\mathcal{O}})\hat{\psi}_p^{\Downarrow} - 2i\frac{\sin\frac{\mathcal{O}}{2}\chi t}{\mathcal{O}}\hat{S}_+\hat{\psi}_p^{\Uparrow}]. \end{aligned} \tag{S38}$$

We now define the operator $\mathcal{O}$ as:

$$\mathcal{O} = \sqrt{4\hat{S}_+\hat{S}_- + (2\hat{S}_Z - 1 + 2\frac{kp}{M\chi})^2} = \sqrt{4\hat{S}^2 + 1 + 4(2\hat{S}_Z-1)\frac{kp}{M\chi} + 4(\frac{kp}{M\chi})^2}. \tag{S39}$$

Finally, we evaluate the action of $\hat{\psi}_p^{\Downarrow}$ in the Heisenberg picture:

$$\begin{aligned}\hat{\psi}_p^{\Downarrow}(t)\left|\frac{N}{2}-m,\frac{N}{2}+m\right\rangle &= e^{-i\frac{(2\hat{S}_Z-1)}{2}\chi t}[(\cos\frac{\mathcal{O}}{2}\chi t+i(2\hat{S}_Z-1)\frac{\sin\frac{\mathcal{O}}{2}\chi t}{\mathcal{O}})\hat{\psi}_p^{\Downarrow}-2i\frac{\sin\frac{\mathcal{O}}{2}\chi t}{\mathcal{O}}\hat{S}_+\hat{\psi}_p^{\Uparrow}]\left|\frac{N}{2}-m,\frac{N}{2}+m\right\rangle \\ &= \phi(p)e^{-i\frac{(2\hat{S}_Z-1)}{2}\chi t}[(\cos\frac{\tilde{\mathcal{O}}\chi t}{2}+2mi\frac{\sin\frac{\tilde{\mathcal{O}}\chi t}{2}}{\tilde{\mathcal{O}}})-2i\frac{\sin\frac{\tilde{\mathcal{O}}\chi t}{2}}{\tilde{\mathcal{O}}}(\frac{N}{2}+m)]\sqrt{\frac{N}{2}-m}\left|\frac{N}{2}-m-1,\frac{N}{2}+m\right\rangle \\ &= \phi(p)e^{-i\frac{(2\hat{S}_Z-1)}{2}\chi t}[\cos\frac{\tilde{\mathcal{O}}\chi t}{2}-i\frac{N}{\tilde{\mathcal{O}}}\frac{\sin\frac{\tilde{\mathcal{O}}\chi t}{2}}{\tilde{\mathcal{O}}}]\sqrt{\frac{N}{2}-m}\left|\frac{N}{2}-m-1,\frac{N}{2}+m\right\rangle \\ &\approx \phi(p)e^{-im\chi t}e^{-i\frac{\tilde{\mathcal{O}}\chi t}{2}}\sqrt{\frac{N}{2}-m}\left|\frac{N}{2}-m-1,\frac{N}{2}+m\right\rangle\end{aligned} \tag{S40}$$

Here, $\tilde{\mathcal{O}}$ is a real number satisfying $\mathcal{O}\left|\frac{N}{2}-m-1,\frac{N}{2}+m\right\rangle=\tilde{\mathcal{O}}\left|\frac{N}{2}-m-1,\frac{N}{2}+m\right\rangle$ and

$$\tilde{\mathcal{O}}=\sqrt{N^2+4(\frac{kp}{M\chi})^2+8m\frac{kp}{M\chi}}\approx N[1+2(\frac{kp}{M\chi N})^2+4m\frac{kp}{M\chi N^2}]. \tag{S41}$$

We then obtain:

$$\boxed{\hat{\psi}_p^{\Downarrow}(t)\left|\frac{N}{2}-m,\frac{N}{2}+m\right\rangle\approx\phi(p)e^{-i\frac{N\chi t}{2}}e^{-im\chi t(1+\frac{2kp}{M\chi N})}e^{-i\chi Nt(\frac{kp}{M\chi N})^2}\sqrt{\frac{N}{2}-m}\left|\frac{N}{2}-m-1,\frac{N}{2}+m\right\rangle}. \tag{S42}$$

Following the same procedure, we also find:

$$\boxed{\hat{\psi}_p^{\Uparrow}(t)\left|\frac{N}{2}-m,\frac{N}{2}+m\right\rangle\approx\phi(p)e^{-i\frac{N\chi t}{2}}e^{i(m-1)\chi t(1+\frac{2kp}{M\chi N})}e^{-i\chi Nt(\frac{kp}{M\chi N})^2}\sqrt{\frac{N}{2}+m}\left|\frac{N}{2}-m,\frac{N}{2}+m-1\right\rangle}. \tag{S43}$$

Eq. (S42) and (S43) constitute the main results of this section.

C. Wave packet density

Now it becomes straightforward to calculate the density in real space. By neglecting the linear term in p, we obtain the following expression:

$$\langle\psi|_0(\hat{\psi}_p^{\Uparrow})^{\dagger}(t)\hat{\psi}_q^{\Downarrow}(t)|\psi\rangle_0\approx\frac{\phi(p)\phi(q)e^{iv_{pq}t}}{2^N}\sum_m\binom{N}{\frac{N}{2}+m}(\frac{N}{2}-m)e^{-i2m\chi t}=\frac{N\phi(p)\phi(q)e^{iv_{pq}t}}{2}e^{i\chi t}\cos^{N-1}\chi t, \tag{S44}$$

where $v_{pq}=k^2\left(p^2-q^2\right)/M^2\chi N$. Similarly, we find:

$$\langle\psi|_0(\hat{\psi}_p^{\Uparrow})^{\dagger}(t)\hat{\psi}_q^{\Uparrow}(t)|\psi\rangle_0=\langle\psi|_0(\hat{\psi}_p^{\Downarrow})^{\dagger}(t)\hat{\psi}_q^{\Downarrow}(t)|\psi\rangle_0=\frac{N\phi(p)\phi(q)e^{iv_{pq}t}}{2}. \tag{S45}$$

Substituting the above results into Eq. (S27), we obtain:

$$\begin{aligned}&\sum_{pq}\langle\psi|_0(\hat{\psi}_p^{\Downarrow})^{\dagger}(t)\hat{\psi}_q^{\Downarrow}(t)|\psi\rangle_0 e^{-i(p-q)Z/\hbar}e^{i\omega_{pq}t}=\sum_{pq}\langle\psi|_0(\hat{\psi}_p^{\Uparrow})^{\dagger}(t)\hat{\psi}_q^{\Uparrow}(t)|\psi\rangle_0 e^{-i(p-q)Z/\hbar}e^{i\omega_{pq}t}\\ =&\frac{N}{2}\sum_{pq}\phi(p)\phi(q)e^{-i(p-q)Z/\hbar}e^{i(\omega_{pq}+v_{pq})t}=\frac{N}{2}F\left(Z,t\right),\end{aligned} \tag{S46}$$

and

$$\begin{aligned}&\sum_{pq}\langle\psi|_0(\hat{\psi}_p^{\Uparrow})^{\dagger}(t)\hat{\psi}_q^{\Downarrow}(t)|\psi\rangle_0 e^{-i(p-q)Z/\hbar}e^{i(\omega_{pq}+v_{pq}+\omega_z)t}e^{-i2k_cZ}\\ =&\frac{N}{2}e^{-i2k_cZ}e^{i(\chi+\omega_z)t}\sum_{pq}\phi(p)\phi(q)e^{-i(p-q)Z/\hbar}e^{i(\omega_{pq}+v_{pq})t}\cos^{N-1}\chi t=\frac{N}{2}F\left(Z,t\right)e^{-i2k_cZ}e^{i(\chi+\omega_z)t}\cos^{N-1}\chi t.\end{aligned} \tag{S47}$$

To cancel the quadratic phase in p, q arising from ω_{pq} and v_{pq}, the optimal interaction strength is chosen such that:

$$\frac{k_c^2 p^2}{M^2 \chi N} + \frac{p^2}{2M\hbar} = 0 \Rightarrow \hbar\chi_{\text{opt}} = -4\frac{(\hbar k)^2}{2M} = -4E_R, \tag{S48}$$

which is consistent with the mean-field prediction. The total atomic density is then given by:

$$\boxed{n_z(t) = \langle \hat{c}_z^\dagger \hat{c}_z \rangle = F(z,t)\left[1 + \cos\left(2kz - (\chi_{\text{opt}} + \omega_z)t\right) \cos^{N-1} \chi_{\text{opt}} t\right].} \tag{S49}$$

Here $F(z,t) = N \exp\left\{-(z - p_0 t)^2 / \sigma_Z(0)\right\} / \sqrt{2\pi}\sigma_Z(0)$ describes the wave packet propagating with center-of-mass momentum p_0 with no diffusion. Notably, the contrast between the $\Downarrow$ and $\Uparrow$ pseudospin components decays over time as $\cos^{N-1} \chi_{\text{opt}} t$, originating from the entanglement generated by the one-axis twisting Hamiltonian $-\chi \hat{S}_Z^2$, which is absent in the mean-field treatment. When the initial state lies on the equator, this Hamiltonian is known to produce spin squeezing. This opens up an avenue for future exploration of entanglement generation in soliton-based interferometry.

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